

Ratings of affective valence closely track changes in oxygen uptake: Application to high-intensity interval exercise

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ABSTRACT

High-intensity interval training (HIIT) has garnered attention because of the promise that it may confer health benefits with reduced time commitment compared to prescriptions calling for moderate-intensity continuous exercise. However, the value of HIIT for public health hinges on whether the high intensity of such regimens will be acceptable or tolerable. While the dual-mode theory predicts that exercise performed at an intensity exceeding critical power (i.e., as most HIIT protocols) will induce displeasure tied to the severe homeostatic perturbation, skeptics have countered that the empirical basis of the theory is limited to continuous exercise protocols and that the theory may not be applicable to intermittent exercise such as HIIT. Using four HIIT protocols, designed to expend either 80 % or 60 % of the finite work that can be performed above critical power over either 5-min or 3-min intervals, we demonstrate that affective valence closely tracks rapid changes in oxygen uptake. These data illustrate the link of affect to homeostatic perturbations and highlight the potential utility of ratings of affective valence as a tool for monitoring exercise stress and customizing training regimens.

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When asked about the reasons for their lack of regular physical activity, most people report that their main barrier is lack of time (Adeniyi, Anjana, & Weber, 2016; Barnett, Guell, & Ogilvie, 2012; Mathew Joseph, Ramaswamy, & Wang, 2018). Behavioral scientists have warned that these self-reports should not be taken at face value since perceived lack of discretionary time is contradicted by evidence showing that discretionary time, in fact, has been increasing in most countries but taken up by sedentary pursuits, such as media consumption (Ekkekakis, Vazou, Bixby, & Georgiadis, 2016; Rebar et al., 2019; Sturm & Cohen, 2019). Instead, a literature that cuts across the fields of psychology, sociology, and economics suggests that time allocation decisions are associated, to a large extent, with the intrinsic 'process benefits' attached to time use (Juster & Stafford, 1985, p. 4), namely the direct enjoyments or satisfactions from the activities themselves (p. 2; also see Juster, 1985; National

Research Council, 2000). When such models of time-allocation behaviors are applied to exercise, they incorporate the assumption that a higher level of exercise intensity would increase the disutility associated with exercise as a behavioral option due to the fact that exercise performed at high intensity will likely be experienced as unpleasant (Meltzer & Jena, 2010).

Despite data emerging from time-allocation research, researchers in exercise science have continued to assume that lack of time is indeed a crucial barrier to exercise participation. On this basis, a prominent research direction over the past several years has focused on devising time-efficient forms of exercise, in which reductions in exercise duration have entailed increases in intensity in order to preserve comparable levels of efficacy. High-intensity interval training (HIIT) has received extraordinary attention as one such solution because it promises to combine significant beneficial adaptations in cardiorespiratory and metabolic parameters while requiring substantially less time compared to traditionally prescribed, moderate-intensity continuous forms of exercise (Gibala, 2018; Gillen & Gibala, 2018).

Recently, the Advisory Committee for the 2018 federal Physical Activity Guidelines in the United States (Campbell et al., 2019)

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concluded that HIIT can improve insulin sensitivity, blood pressure, and body composition in adults and that these improvements are comparable with those resulting from continuous, moderate-intensity aerobic exercise (p. 1225). While the expert panel pointed out that the willingness and the ability of individuals to adhere to HIIT currently are not well known, it also noted that a scoping review of the psychological responses to HIIT concluded that HIIT may be a viable alternative to moderate-intensity continuous exercise (p. 1226). Indeed, a review of 42 published research articles concluded that extant data on psychological responses to HIIT support the viability of interval exercise as an alternative to continuous exercise (Stork, Banfield, Gibala, & Martin Ginis, 2017, p. 341; for a similar conclusion, see Oliveira, Santos, Kilpatrick, Pires, & Deslandes, 2018).

Assertions that HIIT represents a viable alternative to moderate-intensity exercise interventions are being challenged by emerging evidence from pragmatic randomized controlled trials. For example, some trials have found that the intensity had to be lowered for participants to be able to complete their HIIT sessions or that participants spontaneously chose to adopt exercise intensities lower than those prescribed (Conraads et al., 2015; Ellingsen et al., 2017). Other trials have found rapidly declining adherence both during and following the initial HIIT intervention period (Lunt et al., 2014; Roy et al., 2018).

Questions regarding the tolerability of HIIT and the willingness of individuals to partake in it have been examined primarily in studies examining affective (e.g., pleasure-displeasure) responses to HIIT sessions (Oliveira et al., 2018; Stork et al., 2017). Most such studies have been based on the dual-mode theory (Ekkekakis & Brand, 2019). This theory draws insights from affective psychology and neuroscience, scientific fields in which affective experiences of pleasure and displeasure are considered direct reflections of the homeostatic equilibrium of the body (Panksepp, 1998, p. 164; also see Feldman Barrett, 2015, p. 60). Therefore, for the near-maximal exercise intensities typically involved in HIIT, the dual-mode theory predicts homogeneously negative affective responses. At intensities within the so-called severe domain (i.e., beyond the level of critical power, CP; see Poole, Burnley, Vanhatalo, Rossiter, & Jones, 2016) and until the point of peak oxygen consumption ($\dot{V}O_{2\text{ peak}}$), the maintenance of physiological steady state is no longer possible; physiological parameters, such as oxygen consumption and blood lactate, rise continuously until exercise is terminated. In other words, exercise in the severe range is metabolically unsustainable and must be terminated (or the intensity must be appropriately regulated) in order to prevent an impending metabolic crisis. According to dual-mode theory, this is accomplished by a surge of displeasure that compels the exerciser to stop. Moreover, the dual-mode theory predicts that, at this level of intensity, affective responses will closely reflect the severity of homeostatic perturbation, becoming resistant to deliberate efforts to cognitively regulate the displeasure. Evidence produced to date has supported these tenets of the theory (Ekkekakis, Parfitt, & Petruzzello, 2011). In turn, emerging evidence suggests that experiences of displeasure during exercise are associated with reduced subsequent participation in exercise and physical activity (Ekkekakis & Dafermos, 2012; Rhodes & Kates, 2015).

Some authors have questioned whether the predictions of dual-mode theory regarding the association of high exercise intensity with displeasure would apply to interval exercise, pointing out that the theory was developed on the basis of studies involving continuous exercise protocols (Biddle & Batterham, 2015; Jung, Little, & Batterham, 2016; Stork et al., 2017). According to Stork et al. (2017), of those studies that have included assessments of affect during sessions of HIIT and moderate-intensity continuous exercise, four have found that affect was less positive or more negative during

HIIT and four have found no differences between the two. Of the possible reasons for the discrepant findings (for a more complete list, see Decker & Ekkekakis, 2017), two may be especially salient. First, there is great variability in the types of exercise sessions that have been characterized as HIIT in terms of the intensity of the high-intensity intervals, the intensity of the recovery periods, the duration of the intervals, and the ratio of high-intensity intervals to recovery periods (Stork et al., 2017). In most cases, the intensity of the high-intensity intervals, as well as the interval-to-recovery ratios, are selected without any reference to a physiological rationale that could justify, for example, whether the amount of work to be performed per interval can be expected to be metabolically feasible or whether the recovery periods suffice to allow sufficient clearance of metabolic byproducts and/or sufficient restitution of energy resources. Consequently, in several cases, researchers have reported either a surprisingly high number of participants stopping before completing the session (e.g., Conraads et al., 2015; Oliveira, Slama, Deslandes, Furtado, & Santos, 2013) or a variety of adverse events such as lightheadedness, migraines, or vomiting (e.g., Saanijoki et al., 2018).

Second, a frequently overlooked but crucial methodological issue pertains to the timing protocol for sampling ratings of affect from participants. In principle, affect must be sampled with adequate temporal resolution to enable a faithful representation of the shape of the affective response (Ekkekakis, Ladwig, & Hartman, 2019). This point is especially relevant when studying affective responses to HIIT protocols. If participants feel increasingly worse during the high-intensity intervals and progressively better during the recovery periods, in order to faithfully represent this type of oscillating affective response, ratings of affect must be obtained (at least) at the end of each high-intensity interval and at the end of each recovery period. On the other hand, if affect is assessed after the end of each high-intensity interval, measurement would capture the positive rebound (i.e., the relief from the high-intensity stimulus), would miss the preceding precipitous decline, and would thus misrepresent the true shape of the affective response. For example, Stork, Gibala, and Martin Ginis (2018) collected affect ratings during moderate-intensity continuous exercise while the exercise was ongoing, whereas the affect ratings during HIIT were obtained immediately following each of the work bouts and during the last ~20 s of the rest periods (p. 2113, emphasis added).

The purpose of the present study was to examine whether the negative affective changes in response to exercise performed within the severe range of intensity (i.e., above the CP), as predicted by the dual-mode theory (Ekkekakis & Brand, 2019), are observable in HIIT protocols. In doing so, we addressed both aforementioned methodological issues. First, we determined the intensity of HIIT intervals on the basis of a specific physiological parameter, namely the CP, which has been characterized as the genuine boundary separating exercise in which physiological homeostasis can be maintained from exercise in which it cannot (Jones, Burnley, Black, Poole, & Vanhatalo, 2019, p. 1). The CP has been proposed as a conceptually appropriate basis for designing HIIT protocols (Clark, West, Reynolds, Murray, & Pettitt, 2013; Jones & Vanhatalo, 2017; Pettitt, 2016). Specifically, to test the robustness of the hypothesized patterns of affective responses, we tested four HIIT protocols. The workloads were designed in reference to W' , namely the finite work capacity that can be performed above CP (within the severe domain of exercise intensity), such that the high-intensity intervals were designed to deplete 60 % or 80 % of W' in 3 min or 5 min. Thus, the experimental conditions involved four types of high-intensity intervals: (1) 5-min 80 % W' , (2) 5-min 60 % W' , (3) 3-min 80 % W' , and (4) 3-min 60 % W' . Because of the appearance of a slow component of oxygen uptake ($\dot{V}O_2$) above the CP, these high-intensity intervals were predicted to gradually raise $\dot{V}O_2$ toward its peak value.

Second, we sampled affect (specifically, affective valence or pleasure-vs-displeasure) with high temporal resolution (once per min). This frequency was selected in order to enable the faithful representation of the shape of the affective response throughout the HIIT sessions (Ekkekakis et al., 2019). A similar approach has been used in studies of incremental exercise tests, during which intensity is raised progressively until volitional termination, and therefore the physiological condition of the body changes rapidly (see Ekkekakis et al., 2011, for a review).

In accordance with the dual-mode theory, we theorized that, in a situation in which the homeostatic balance is perturbed (i.e., during exercise above the CP), affective responses would closely track the physiological condition of the body. Based on the assumption that affect provides a continuous readout of bodily homeostatic information, we hypothesized that affect ratings would not only track gross oscillations in exercise intensity across the interval-recovery cycles but would also mirror changes in $\dot{V}O_2$ during each high-intensity interval.

1. Methods

1.1. Participants

This was a within-subjects experiment with four distinct experimental conditions that involved either three (80 % W') high-intensity intervals interspersed with three recovery periods or four (60 % W') high-intensity intervals interspersed with four recovery periods. An a priori power analysis was conducted assuming repeated-measures Analyses of Variance (ANOVAs) with either seven time points (80 % W' : baseline plus six peaks) or nine time points (60 % W' : baseline plus eight peaks), a large main effect of Time (Cohen's $f=0.40$, $\eta_p^2=0.1379$), 80 % statistical power at $\alpha=0.05$, measurement that reflects at least 50 % true variance (i.e., test-retest correlation coefficients of ≥ 0.70), and a violation of sphericity ($\epsilon=.70$). This analysis indicated that 6 participants were needed ($1-\beta=0.81$ for seven repeated measures, $1-\beta=0.89$ for nine repeated measures). To account for attrition, eight individuals were recruited through flyers that identified the study as involving interval exercise (six men, two women, age = 27.75 ± 7.54 years, height = 176.38 ± 9.63 cm, body mass = 73.84 ± 11.47 kg, $\dot{V}O_{2\text{ peak}} = 58.71 \pm 6.86$ ml·kg⁻¹·min⁻¹). All participants attended all sessions. To minimize risk and maximize compliance with the strenuous exercise stimuli, participant recruitment targeted athletic young adults from university cycling and triathlon clubs, all of whom had extensive cycling experience. Written informed consent was obtained prior to the study. A university institutional review board for the protection of human participants preapproved all procedures Approval: IRB Net #815129.

1.2. Measure

Ratings of affective valence (pleasure-displeasure) were obtained using the Feeling Scale (FS; Hardy & Rejeski, 1989) every minute during exercise. The FS is an 11-point, single-item, bipolar rating scale developed for the assessment of affective valence during exercise. The scale ranges from +5 to -5. Anchors are provided at zero (Neutral) and all odd integers, ranging from Very good(+5) to Very bad(-5). Hardy and Rejeski (1989) have provided evidence of significant correlations between the FS and other self-report measures of pleasure.

1.3. Procedures

Participation in this within-subjects experimental study entailed five visits to the laboratory. The first visit included a 3-min all-out exercise test (3 MT) for the estimation of the CP. The

subsequent four visits, each lasting a total of approximately 45 min, were the experimental HIIT sessions. They were performed in counterbalanced order, to avoid an order effect: (1) 5-min 80 % W' , (2) 5-min 60 % W' , (3) 3-min 80 % W' , and (4) 3-min 60 % W' .

All five visits were completed within a period of less than two weeks for all participants. The four experimental sessions were completed at least 48 h from each other (to allow adequate recovery) but less than one week from each other (to prevent any substantive changes in cardiorespiratory fitness in the interim). All participants were instructed to abstain (and self-reported abstaining) from alcohol and heavy exercise for 24 h prior to each visit.

Each participant was instructed to bring his or her own road bicycle to the laboratory for every testing session. The bicycles were affixed to a CompuTrainer (Racermate, Seattle, WA) using the bolts on the rear axle. The CompuTrainer was calibrated according to manufacturer guidelines. Because cadence naturally fluctuates during cycling, the CompuTrainer automatically adjusts the load to keep the power output constant. Power and cadence were sampled at 1 Hz via an analog-to-digital converter on a laptop computer linked to the CompuTrainer. Participants were instructed to try to maintain constant cadence during the exercise intervals. To assist them, cadence (but neither power nor time) was shown on a 70-inch monitor placed in front of them.

During all exercise conditions, breath-by-breath expired gases were analyzed with a metabolic cart (Parvomedics, Logan, UT, USA). Participants were fitted with a nose clip and expired through a two-way non-rebreathing valve connected to the metabolic cart. Filter replacement and calibration of the pneumotach and O_2 and CO_2 sensors were performed between tests, in accordance with guidelines provided by the manufacturer. Gas exchange data were converted offline to 15 s averages for subsequent analysis.

1.3.1. First visit: 3-minute all-out cycling test (3MT)

The 3MT (Burnley, Doust, & Vanhatalo, 2006; Vanhatalo, Doust, & Burnley, 2007) was administered during the initial visit, to estimate CP and W' , using a protocol specifically validated for the CompuTrainer (Clark, Gartner, Williams, & Pettitt, 2016). Following a 10-min warm-up, participants were instructed to pedal at an all-out pace for 180 s (changing gears was allowed). After the 3 MT, participants cycled at 50 W for a 3-min active recovery period. CP was estimated as the average power output during the last 30 s of the 3 MT, and W' was estimated by the formula $W' = 150 \times (P_{150s} - CP)$, where P_{150s} is the average power for the first 150 s of the 3 MT.

Following recovery, participants performed a verification bout of constant-load exercise at a workload 10 % above CP until exhaustion, while pedaling at their preferred cadence (changing gears was not allowed). Exhaustion was defined as the point at which participants could no longer sustain their initially preferred cadence, defined as a decline of at least 5 rpm for at least 5 s (Johnson, Sexton, Placek, Murray, & Pettitt, 2011). A test was deemed valid if the highest oxygen uptake elicited during the constant-load bout was equal to or exceeded the highest value of oxygen uptake recorded during the 3 MT. The highest oxygen uptake observed during the constant-load bout was designated as $\dot{V}O_{2\text{ peak}}$ (Clark et al., 2016; Clark, Murray, & Pettitt, 2013; Dicks, Jamnick, Murray, & Pettitt, 2016).

1.3.2. HIIT bouts

On visits two through five, participants completed the four experimental sessions of high-intensity interval exercise. The cadence during the intervals was the same as the preferred cadence used in the verification bout performed subsequent to the 3 MT (changing gears was not allowed). Participants warmed up on their own bicycles for 10 min while the CompuTrainer was calibrated. Power outputs for the intervals were prescribed by the equation $P = (\%W'/t_{lim}) + CP$, where P is the power output for the intervals, % W'

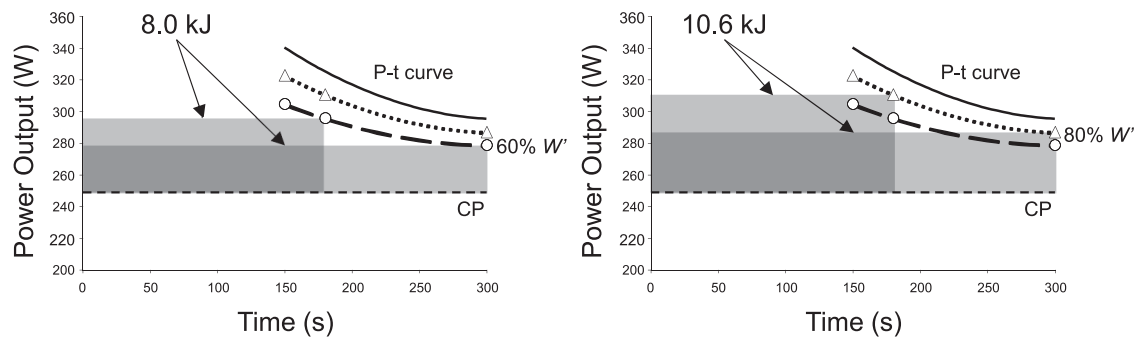


Fig. 1. Illustration of the high-intensity interval training prescriptions using the critical power (CP) concept (data from a representative participant). The participant's CP of 252 W and W' of 13.3 kJ were used to model the estimated power (P-t curve, solid black lines) at t_{lim} of 150, 180, and 300 s using the equation $P = W'/t_{lim} + CP$ (see Methods). The open triangles and open circles represent curves for 80 % and 60 % of W' , respectively. The fractional utilization of W' for 3-min and 5-min intervals are shown with rectangles (of equal area) for 80 % and 60 %.

is the fractional depletion of the W' (0.60 or 0.80) and t_{lim} is the time of each interval in s (180 s or 300 s). Both the 60 % and 80 % percentages of W' have been found to be sustainable over 3 min and 5 min intervals for running (Clark, Murray et al., 2013), cycling (Pettitt, Placek, Clark, Jamnick, & Murray, 2015), and swimming (Courtright, Williams, Clark, Pettitt, & Dicks, 2016).

For the 60 % W' depletion trials, the participants completed four intervals, whereas for the 80 % W' depletion trials, they completed three intervals because more would have been physiologically untenable. Fig. 1 illustrates how these intervals were derived using data from a representative participant. Active recovery periods at a workload equal to 50 % of CP and of duration equal to the duration of the exercise intervals (i.e., 3 min or 5 min, respectively) were used between intervals, to allow for standardized restitution of W' . FS ratings were obtained 10 s prior to the completion of each min during each interval and recovery period, by asking participants to point to a rating on a poster-size copy of the FS (approximately 28 by 43 cm).

1.3.3. Statistical analyses

Based on evidence that certain features of the affective response carry disproportionate weight in shaping memories of events (Ekkekakis, Zenko, Ladwig, & Hartman, 2018; Zenko, Ekkekakis, & Ariely, 2016), while all FS assessments were retained for descriptive purposes (totaling 19, 25, 31, and 41 FS ratings; see Fig. 2), for the purposes of statistical analysis we extracted (a) the positive and negative peaks (highest and lowest values), and (b) the slopes of FS observed during the high-intensity intervals and recovery periods. Repeated-measures ANOVAs for the conditions involving 80 % W' (three intervals and recovery periods) included seven time points (baseline, first interval, first recovery, second interval, second recovery, third interval, third recovery), whereas ANOVAs for the conditions involving 60 % W' (four intervals and recovery periods) included nine time points (with the addition of the fourth interval and fourth recovery). When violations of the assumption of sphericity occurred, the degrees of freedom were adjusted using the Greenhouse-Geisser method (indicated by decimal degrees of freedom). In the case of significant effects, follow-up t -tests were subject to Bonferroni adjustment to prevent the inflation of the Type I error rate due to the multiple comparisons. In such cases, the reported probability values represent the observed (uncorrected) p -value multiplied by the number of comparisons. Therefore, for convenience, these p values can be judged by the conventional criterion of $p < .05$.

2. Results

2.1. Manipulation checks

CP averaged 249.4 ± 31.6 W and W' averaged 12.7 ± 3.6 kJ. Individualized CP and W' parameters were used to establish 3-min and 5-min intervals with 60 % and 80 % W' , respectively. The power outputs associated with each of the four experimental conditions differed significantly, $F(1.007, 7.048) = 99.022$, $p < .0001$, $\eta^2 = .93$, with all pairwise comparisons between conditions being significant (for all, $p < .001$). On the other hand, while a repeated-measures ANOVA on the $\dot{V}O_{2\text{ peak}}$ values derived from each of the four experimental conditions and the initial visit was significant, $F(3.379, 23.653) = 3.498$, $p = .027$, $\eta^2 = .33$, none of the Bonferroni-corrected pairwise comparisons was significant. Participants reached 92 % $\dot{V}O_{2\text{ peak}}$ during the 5-min 80 % W' session, 91 % $\dot{V}O_{2\text{ peak}}$ during the 5-min 60 % W' session, 96 % $\dot{V}O_{2\text{ peak}}$ during the 3-min 80 % W' session, and 95 % $\dot{V}O_{2\text{ peak}}$ during the 3-min 60 % W' session.

Out of 32 experimental sessions (4 for each of the 8 participants), there were 6 discontinuations: (1) 5-min 80 % W' : 1 discontinuation, with final FS rating of -5; (2) 5-min 60 % W' : 2 discontinuations, with final FS ratings of -4 for both; (3) 3-min 80 % W' : 2 discontinuations, with final FS ratings of -2 and -5; (4) 3-min 60 % W' : 1 discontinuation, with final FS rating of -4.

2.2. Changes in affective valence

FS ratings during all four experimental conditions changed significantly over time (see Fig. 2): (1) 5-min 80 % W' : $F(1.99, 11.94) = 19.29$, $p < .001$, $\eta^2 = .76$; (2) 5-min 60 % W' : $F(8, 40) = 12.26$, $p < .001$, $\eta^2 = .71$; (3) 3-min 80 % W' : $F(6, 30) = 22.92$, $p < .001$, $\eta^2 = .82$; (4) 3-min 60 % W' : $F(8, 48) = 11.90$, $p < .001$, $\eta^2 = .67$. In each condition, the final FS rating of the final interval was significantly lower than the baseline rating after Bonferroni correction: (1) 5-min 80 % W' : $p = .007$, $d = 2.77$; (2) 5-min 60 % W' : $p < .001$, $d = 8.10$; (3) 3-min 80 % W' : $p = .035$, $d = 2.50$; (4) 3-min 60 % W' : $p = .050$, $d = 2.11$. On the other hand, none of the final FS ratings of the final recovery periods was significantly different from the baseline rating after Bonferroni correction (for all comparisons, $p = 1.00$; $d = 0.10$, $d = 0.37$, $d = -0.39$, $d = 0.16$, respectively).

Repeated-measures ANOVAs examining the change in the negative peaks in FS from the first to the last high-intensity interval showed that the negative peaks grew progressively larger over the course of the sessions, with the exception of the 3-min 80 % W' condition: (1) 5-min 80 % W' : $F(1.308, 7.849) = 5.027$, $p = .049$, $\eta^2 = .46$ (first: -0.286 ± 2.498 ; second: -1.571 ± 2.299 ; third: -2.714 ± 2.498); (2) 5-min 60 % W' : $F(1, 817, 10.900) = 12.100$, $p = .002$, $\eta^2 = .67$ (first: 1.714 ± 1.799 ; second: 0.714 ± 3.039 ;

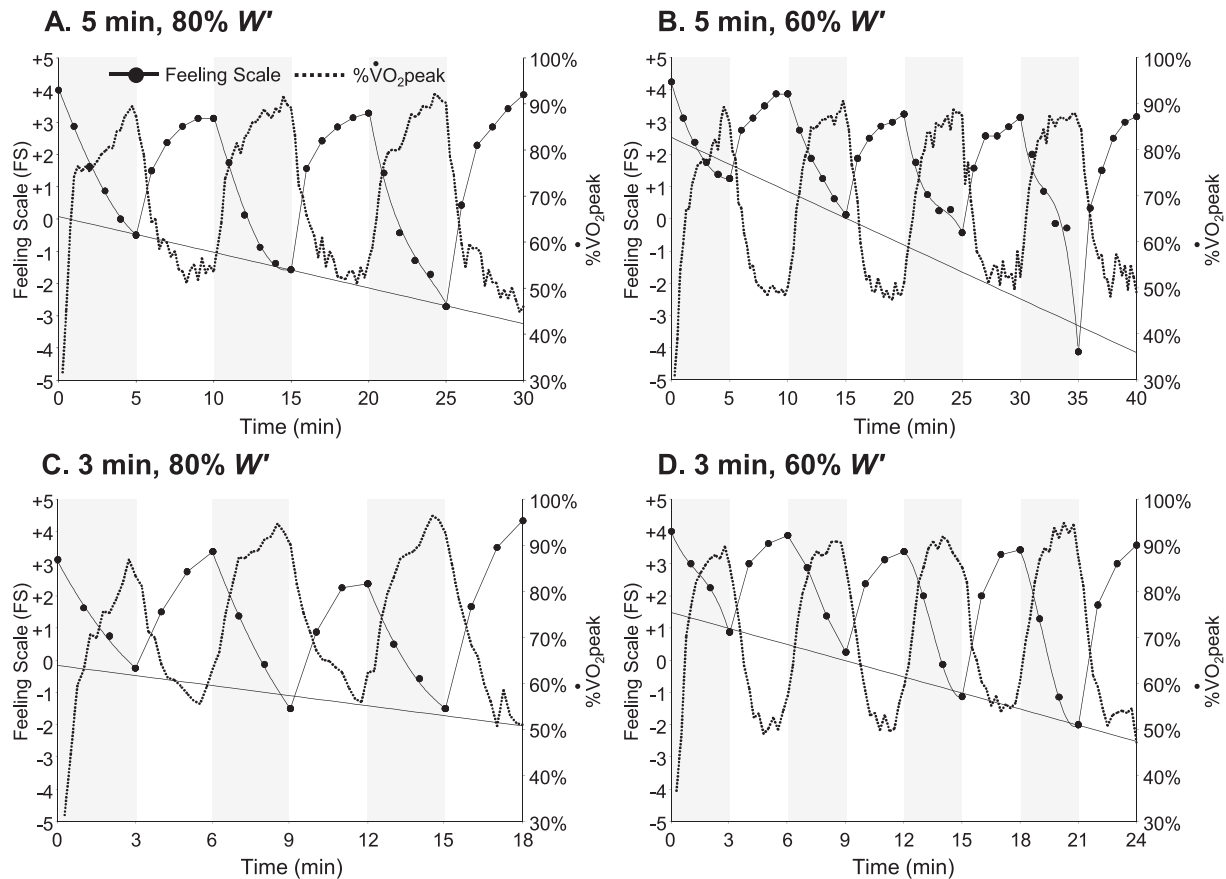


Fig. 2. Averaged min-by-min FS ratings, coregistered with percentages of peak oxygen uptake ($\% \dot{V}O_{2\text{peak}}$) during the four experimental conditions. The linear slopes shown near the bottom of each panel represent the lines of best fit across the negative affective peaks (i.e., the FS ratings obtained before the end of each high-intensity interval), illustrating that the peaks became increasingly more negative across the high-intensity intervals.

third: -0.429 ± 3.910 ; fourth: -4.143 ± 0.378); (3) 3-min 80 % W': $F(1.502, 7.509) = 2.699$, $p = .137$, $\eta^2 = .35$ (first: 0.000 ± 1.673 ; second: -0.833 ± 2.317 ; third: -1.500 ± 1.871); (4) 3-min 60 % W': $F(1.781, 10.684) = 4.544$, $p = .041$, $\eta^2 = .43$ (first: 1.000 ± 2.160 ; second: 0.571 ± 2.440 ; third: -0.714 ± 1.604 ; fourth: -2.000 ± 2.582). However, the lowest peak in FS ratings (in all cases, the peak associated with the last high-intensity interval) did not differ significantly between experimental conditions, $F(1.921, 13.450) = 1.646$, $p = .229$, $\eta^2 = .19$.

A condition by interval (first, last) ANOVA on the negative slopes of FS ratings during the exercise intervals showed only a significant main effect of interval, $F(1, 7) = 41.332$, $p < .001$, $\eta^2 = .86$, whereas the main effect of experimental condition, $F(1.654, 11.579) = 1.827$, $p = 0.206$, $\eta^2 = .21$, and the condition by interval interaction, $F(1.626, 11.381) = 0.915$, $p = .408$, $\eta^2 = .12$, were not significant. The difference from the first (-0.905) to last (-1.469) slopes, averaged across the four conditions, was significant ($t = 6.429$, $p < .001$).

3. Discussion

The purpose of the present study was to provide empirical evidence regarding predictions derived from the dual-mode theory (Ekkekakis & Brand, 2019) on affective responses to exercise performed in the severe domain of intensity (i.e., above the CP). The dual-mode theory predicts that, in the severe range of exercise intensity, at which homeostasis is significantly perturbed, affect will closely track changes in the physiological condition of the body, becoming more negative as the severity of the homeostatic perturbation intensifies. Because core affect is theorized to be a

component of consciousness inherently tied to homeostatic regulation (Feldman Barrett, 2015; Panksepp, 1998), we posited that this prediction applies to all types of exercise, regardless of whether the format is continuous or intermittent. As shown in Fig. 2, changes in ratings of affective valence during the four experimental conditions mirrored the dynamics of $\dot{V}O_2$, responding to increases and decreases in instantaneous and proportional fashion. Thus, the present findings support the notion that the predictions of the dual-mode theory hold in the case of interval exercise, as they do in the case of continuous exercise (Ekkekakis et al., 2011; Hartman, Ekkekakis, Dicks, & Pettitt, 2019).

From a practical standpoint, these findings highlight the potential utility of ratings of affective valence in monitoring the internal load of exercise, namely the response of the body to various exercise stimuli (in contradistinction to external load, such as mechanical measures of work). Indeed, the American College of Sports Medicine (2018), in its guidelines for exercise testing and prescription, has recognized that measures of . . . affective valence (i.e., the pleasantness of exercise) can be used to modulate or refine the prescribed exercise intensity (p. 151). Based on the present results, this recommendation could be extended to interval forms of exercise in addition to continuous exercise.

An interesting observation also made by others (e.g., Decker & Ekkekakis, 2017; Frazão et al., 2016; Stork et al., 2018) is that the negative peaks that occur at the end of each high-intensity interval tend to become increasingly larger with additional intervals. In the present study, the negative slopes of FS ratings during the last high-intensity interval were larger than the negative slopes of the first intervals. Moreover, in three of the four experimental conditions, there was a significant stepwise decline in the negative

peaks across intervals. These findings may have important motivational implications since, as noted earlier, both the amplitude of the unpleasant peaks and the slope of affective valence over the course of an episode have been found to weigh heavily on how the episode registers in memory (Ekkekakis et al., 2018; Zenko et al., 2016). For example, Zenko et al. (2016) reported that the slope of FS ratings over the course of a 15-min bout of recumbent cycling accounted for 35–46 % of the variance in remembered and forecasted pleasure from 15 min to 7 days postexercise.

The phenomenon of the increasingly more negative peaks in affective valence possibly reflects a combination of interoceptive factors beyond $\dot{V}O_2$, such as the inadequate restitution of high-energy phosphate and muscle glycogen stores, and the incomplete clearance of metabolic byproducts, such as lactic acid. This highlights a persistent challenge in HIIT protocols (Schoenmakers, Hettinga, & Reed, 2019), as there is no easy way to assess these factors and it appears that general rules of thumb (e.g., a recovery interval of 3 min is expected to be sufficient to avoid premature fatigue, according to Schoenmakers et al., 2019) are bound to fail in individual cases. As a case in point, despite the careful individualization of workloads in the present study, as well as the use of recovery periods with durations of ≥ 3 min in a 1:1 ratio, there were several discontinuations. The fact that most of these discontinuations occurred with final FS ratings of -4 or -5 underscores the potential of using the FS as a practical tool in gauging the internal and otherwise unobservable physiological state of the organism (e.g., the degree of replenishment of energy resources or degree of clearance of metabolic byproducts) and adjusting recovery periods accordingly on a flexible, rather than fixed, schedule.

From a methodological standpoint, the results of the present study reemphasize the need for researchers, peer reviewers, and journal editors to insist on standardizing affect-assessment protocols across experimental conditions and, in particular, the need for sampling to capture the true shape of the affective response over time (Ekkekakis & Brand, 2019; Ekkekakis et al., 2019). The data shown here demonstrate the dynamic nature of affective responses and especially the instantaneous reversals when the intensity is raised (and affective valence starts to decline) and when the intensity is lowered (and affective valence starts to improve). By extension, expert panels and guideline developers must scrutinize claims that affective responses to HIIT are comparable to those that occur during moderate-intensity continuous exercise (e.g., Oliveira et al., 2018; Stork et al., 2017) because, in many studies comparing the two modes, affective responses were assessed under different conditions (i.e., while moderate-intensity continuous exercise was still ongoing vs. after high-intensity intervals had ended).

While the present study had several strengths (i.e., theoretical grounding, a physiological rationale for the determination of exercise loads, four HIIT protocols to test the robustness of the results), it also had weaknesses that critical readers should consider. First, in order to reduce the likelihood of adverse events given the strenuous nature of the exercise stimuli, the participants were young, physically fit, lean, and athletic. Therefore, the findings should not be assumed to generalize to populations with different characteristics. It is possible, for example, that the declines in affective valence reported here might have been even steeper, and the positive rebounds during the recovery periods might have been relatively delayed or diminished, in a sample of participants not as young, fit, lean, or athletic. Likewise, the patterns of affective responses reported here cannot be assumed to mimic those that occur in response to other HIIT protocols, such as the commonly employed set of 1-min intervals interspersed with 1-min recovery periods (e.g., Stork et al., 2017). While the basic oscillatory pattern has been extensively replicated and appears robust across different HIIT protocols, the amplitude of the positive and negative peaks may be attenuated with shorter intervals (e.g., Stork et al., 2018).

While the valleys may not be as low (at least during the early intervals) as those found here in response to 3-min and 5-min intervals, the positive rebounds during the recovery periods may also not be as high, presumably due to the 1-min recovery being too short to allow for adequate restitution of energy resources and clearance of metabolic byproducts. Thus, in a sample of young but low-fit participants, Stork et al. (2018) found that FS ratings remained below zero during the eighth and ninth (out of 10) recovery periods.

Finally, in this study, we employed assessments of affect with high temporal resolution (once per min) as we deemed this necessary to provide a fine-grained representation of the affective response to HIIT (Ekkekakis et al., 2019). However, this frequency of sampling should be used judiciously in research due to the possibility of artifacts, including reactivity to repeated assessments, forced introspection or internal attentional focus, or response carry-over effects. Likewise, in professional practice, while we encourage exercise professionals to obtain ratings at least at the lowest part of the valleys (i.e., the end of the high-intensity intervals) and the highest part of the peaks (i.e., the end of recovery periods), a higher frequency of assessment may be unnecessary for monitoring and customization purposes.

In summary, in this study, we demonstrated that ratings of affective valence acquired during four HIIT protocols closely tracked rapid changes in oxygen uptake. From a theoretical standpoint, these findings provide evidence that the prediction of the dual-mode theory regarding declines in affective valence in response to severe-intensity exercise (above CP) also applies to interval exercise. From an applied standpoint, these data highlight the potential utility of ratings of affective valence as a tool that can help exercise practitioners obtain a summary index of internal loads not otherwise accessible by routine monitoring in most field and clinical settings.

Declaration of Competing Interest

None.

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